PRE-ALGEBRA (7TH) · UNIT 9: SPATIAL THINKING

Skate Park Designer

Use angles, polygons, circles, and transformations to design a town skate park.

CP1

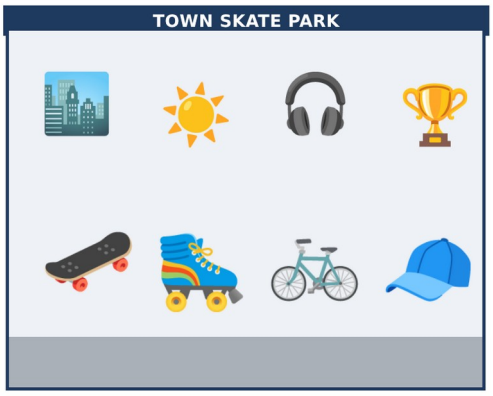
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The town council approved a new skate park and wants students to design it. The ramps need exact angles, the rails run parallel, the deck is a polygon, and the bowl is a circle.

You will use angle relationships, polygons, circles, and transformations to create a blueprint and a logo for the park.



Ramp Rule
Angle of ramp
15° to 35°

Bowl
Circle, diameter
24 ft

Learning Goals

- I can use angle pair relationships to find unknown angles.
- I can use parallel lines, polygon angle sums, and congruence in design.
- I can apply circle formulas and transformations.

Project Timeline

When	What you will do
Day 1	Ramp Geometry
Day 2	Layout
Day 3	Bowl & Logo
Final	Skate Park Blueprint + reflection

Part 1: Ramp Geometry

Day 1 · Use the Feature Cards.

Feature Cards

Quarter Pipe	Fun Box	Rail Pair
Ramp angle 30°	Regular hexagon top	Two parallel rails cut by a path

Bowl	Pyramid	Logo
Circle, d = 24 ft	Square base, 4 triangle sides	Must have rotational symmetry

1. A ramp and its support form a right triangle with the ground. If the ramp angle is (2x + 4)° and the angle at the top is (3x + 1)°, find x and both angles.

2. Does the 38° ramp meet the rule (15° to 35°)? What angle would you change it to?

3. Two crossing rails form angles (4y − 10)° and (2y + 50)° that are vertical. Find y and the angle.

4. Explain why vertical angles are always equal.

Part 2: Layout

Day 2 · Two parallel rails are crossed by a path.

5. An alternate interior angle is (5x − 20)° and its partner is (3x + 30)°. Find x and the angle.

6. The fun box top is a regular hexagon and the pyramid base is a square. Find each interior angle and the sum of the exterior angles of each.

7. Two ramp triangles have matching sides 6 ft, 8 ft, 10 ft. Are they congruent? Which rule?

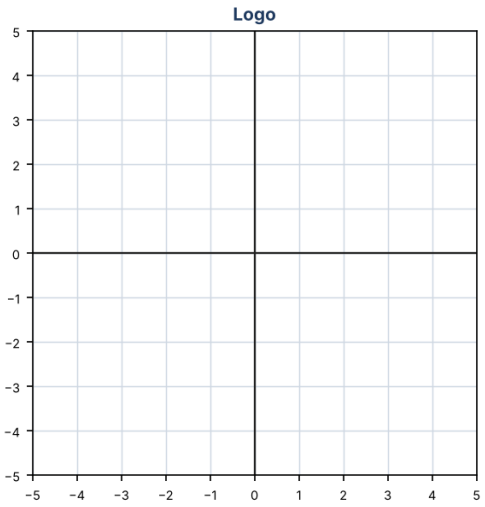
8. What polygon has an interior angle sum of 1,080°? Could it be the deck?

Part 3: Bowl & Logo

Day 3 · Use π ≈ 3.14.

9. Find the circumference and area of the 24-ft bowl. A coping rail goes around the edge at $12 per foot. Find its cost.

10. The logo is a triangle with vertices (1, 1), (4, 1), (1, 3). Rotate it 90° counterclockwise about the origin. Give the new vertices.



11. Design a logo with rotational symmetry of order 3 or more. Give its angle of rotation.

Final Product: Skate Park Blueprint

Your final product must include:

- Blueprint with every angle labeled and justified
- Equations for unknown angles
- Polygon angle calculations
- Bowl measurements and coping cost
- Logo with a rotation and its symmetry

Reflection

R1. How did writing equations help you find angles?

R2. How is a rotation different from a translation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Angles, polygons, circle measures, and transformations are correct and produce a buildable design. Goes beyond the minimum with a creative, realistic choice.	Angles, polygons, circle measures, and transformations are correct and produce a buildable design.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Blueprint is labeled with angles and measurements and includes the logo and calculations. Polished, organized, and visually engaging.	Blueprint is labeled with angles and measurements and includes the logo and calculations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.